Solve One-Step Addition and Subtraction Equations

Lesson 7-2

Name: ______________ Date: ______________ Class: ______________

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

$x + 5 = 12$ — both sides equal 12 when $x = 7$

Equation

A math sentence with an equal sign showing both sides are the same.

Addition undoes subtraction: if $x - 3 = 10$, add 3 to get $x = 13$

Inverse operation

Two math actions that undo each other, like × and ÷.

$x + 5 = 12$ → subtract 5 from both sides → $x = 7$ (x is isolated)

Isolate

To get the letter by itself on one side.

$x = 7$ is the solution to $x + 5 = 12$ because $7 + 5 = 12$ ✓

Solution

The number that makes the equation true.

$x + 4 = 9$ is solved in one step: subtract 4 from both sides, so $x = 5$.

One-step equation

An equation you can solve in a single step using one inverse operation.

If you subtract 4 from the left side, you must subtract 4 from the right to keep balance.

Balance

Keeping both sides of an equation equal by doing the same thing to each side.

Key Ideas & Notes

🎯 **WHAT WE'RE LEARNING TODAY**

I can solve one-step addition and subtraction equations using inverse operations.

1️⃣ Notice

Detective Chen found a coded message: 'The suspect's locker number plus 23 equals 58.' She needs to figure out the locker number to crack the case.

2️⃣ Key idea

I can solve one-step addition and subtraction equations using inverse operations.

3️⃣ Words I'll use

Equation

Inverse operation

Isolate

Solution

One-step equation

Balance

🤔 Think About It

- What operation connects the locker number and 23?
- What does n represent in the equation?
- How could you undo the addition to find n?

👀 Watch

Watch your teacher model one example. Jot what you see.

🤝 We try

Solve the next one together as a class.

✏️ You try

Now try one on your own.

Turn & Talk — Discussion Points

Level 1 support · Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Detective Chen's clue is n + 23 = 58, where n is a locker number. Why can't she just 'know' n by looking, and what could she do to both sides to find it?

Level 1 support

Start here: I can undo the addition to get the variable by itself.

💡 **Need a hint?**

Sentence starters (optional)

The variable n is added to ____, so to find n I can ____.

La variable n está sumada a ____, así que para hallar n puedo ____.

I would use ____ to undo the ____.

Usaría ____ para deshacer la ____.

Word bank: equation · inverse operation · isolate · solution

Listen for: *Students recognize 23 is added to n and that subtracting 23 (the inverse operation) from both sides will isolate n.*

Level 2

How would your plan change if the clue were n − 23 = 58 instead? Justify which inverse operation you would use.

- For n − 23 = 58 I would ____ instead because ____.
- The inverse operation changes because ____.

EXPLORE

What inverse operation did you use to solve n + 23 = 58, and why must you do it to BOTH sides of the balance?

Level 1 support

Start here: I do the same inverse operation to both sides to keep it balanced.

💡 **Need a hint?**

Sentence starters (optional)

I used ___ to undo ___, so the variable equals ___.

Usé ___ para deshacer ___, así que la variable es igual a ___.

I do it to both sides because ___.

Lo hago en ambos lados porque ___.

Word bank: **inverse operation** **isolate** **solution** **equation**

Listen for: *Students subtract 23 from both sides to get n = 35 and explain that doing the same to both sides keeps the equation balanced and the solution true.*

Level 2

How can you check that n = 35 is the solution? Show why checking matters even when your steps look correct.

- I can check by ___, which gives ___.
- Checking matters because ___.

CONNECT

Where might solving a one-step equation help you find a missing amount in real life?

Level 1 support

Start here: A one-step equation finds an unknown when I know a total and a part.

💡 **Need a hint?**

Sentence starters (optional)

A one-step equation would help me find ___ when I know ___.

Una ecuación de un paso me ayudaría a hallar ___ cuando sé ___.

I would use the inverse operation ___ to solve it.

Usaría la operación inversa ___ para resolverla.

Word bank: equation | inverse operation | isolate | solution

Listen for: *Students give a realistic context (money saved, distance left, missing count) and connect it to isolating a variable using an inverse operation.*

Level 2

Critique: 'Adding the same number to both sides changes the answer.' Use the idea of keeping the equation balanced to explain why this is false.

- This is false because ___.
- Doing the same operation to both sides keeps ___.

Guided Examples

Example 1

Solve: $x + 9 = 15$

Solution: Subtract 9 from both sides: $x = 15 - 9 = 6$. Check: $6 + 9 = 15$ ✓

Answer: A. $x = 6$

Example 2

Solve: $y - 7 = 20$

Solution: Add 7 to both sides: $y = 20 + 7 = 27$. Check: $27 - 7 = 20$ ✓

Answer: A. $y = 27$

Example 3

Solve: $n + 15 = 42$

Solution: Subtract 15 from both sides: $n = 42 - 15 = 27$. Check: $27 + 15 = 42$ ✓

Answer: A. $n = 27$

Write About the Math

The Writing Revolution

I can explain my steps using the words equation, inverse operation, isolate, and solution.

1. Kernel Sentence

subject + verb

Model: Equation is a math sentence with an equal sign showing both sides are the same.
Ecuación es una oración matemática con un signo igual que muestra que ambos lados son iguales.

Write a kernel sentence about equation. Use a subject and a verb.
Escribe una oración base sobre ecuación. Usa un sujeto y un verbo.

____

____

2. Sentence Expansion

because · but · so

Kernel: Equation matters in math
Ecuación importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because *porque*	**Equation matters in math because ____.** *Ecuación importa en matemáticas porque ____.* ______________________________
but *pero*	**Equation matters in math, but ____.** *Ecuación importa en matemáticas, pero ____.* ______________________________
so *entonces*	**Equation matters in math, so ____.** *Ecuación importa en matemáticas, entonces ____.* ______________________________

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about equation.
Di un hecho verdadero sobre equation.

Equation ____.

Question
Pregunta

Ask a question about equation.
Haz una pregunta sobre equation.

How does ____ ?
¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about equation.
Muestra entusiasmo sobre equation.

Wow, ____ !
¡Guau, ____ !

Command
Mandato

Tell a partner what to do with equation.
Dile a un compañero qué hacer con equation.

First, ____ .
Primero, ____ .

4. Explain Your Reasoning

use a sentence starter

The variable n is added to ____, so to find n I can ____.
La variable n está sumada a ____, así que para hallar n puedo ____.

I would use ____ to undo the ____.
Usaría ____ para deshacer la ____.

I used ____ to undo ____, so the variable equals ____.
Usé ____ para deshacer ____, así que la variable es igual a ____.

Try It

Solve on your own. Check the answer key when you are done.

1. Solve: $m - 11 = 25$

A. $m = 36$

B. $m = 14$

C. $m = 25$

D. $m = 275$

Show your work:

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2. A shirt costs \$d. After a \$15 discount, it costs \$38. What is d?

A. \$53

B. \$23

C. \$38

D. \$45

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A detective solved $x + 25 = 60$ and got $x = 35$. Another detective solved $y - 25 = 60$ and got $y = 35$. Are both correct? Explain using inverse operations and check each answer.

Sentence starter: For $x + 25 = 60$, the inverse operation is ____, so $x =$ ____. Check: ____. For $y - 25 = 60$, the inverse operation is ____, so $y =$ ____. Check: ____. Therefore, ____.

Show your work:

Reflect — Exit Ticket

Solve: $p + 2.8 = 9.1$

A. $p = 6.3$

B. $p = 11.9$

C. $p = 6.8$

D. $p = 3.25$

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. m = 36 *— Add 11 to both sides: m = 25 + 11 = 36. Check: 36 − 11 = 25 ✓*
2. **Try It 2:** A. $53 *— d − 15 = 38 → d = 38 + 15 = 53.*
3. **Exit Ticket:** A. p = 6.3 *— Subtract 2.8 from both sides: p = 9.1 − 2.8 = 6.3. Check: 6.3 + 2.8 = 9.1 ✓*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Equation is a math sentence with an equal sign showing both sides are the same.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).